

Checklist for group 7.1 by Daniel Tyukov

Instructions

Write an individual peer review about the draft version of another group. Follow the steps below:

1. Provide feedback on the criteria specified in 2. Checklist (page 2);
 - Label this document as 'Checklist for group x.x by [your name]'
2. If you want, you can also make in-text comments in the article you have to review;
 - Label this document as 'In-text feedback for group x.x by [your name]'
3. Submit your documents twice to Brightspace (see instructions in Assignment 4):
 - a. In Assignments
 - b. In the group locker of your lecture group

When you give peer feedback, please:

- Provide comments on each section of the checklist.
- Clearly state to which part of the text your comments relate.
- If possible, suggest how you think the text could be improved or clearly state what you do not understand about the text.
- Clarify your comments by providing examples from the text.
- Do not focus too much on lower order concerns, such as spelling. If you see that someone has made many spelling errors, please point this out to them and maybe give two or three examples.
- Be respectful but clear in how you phrase the comment.

Tips for effective reviewing

From research by Nelson and Schunn (2007), we know that it is very important for writers to understand the feedback they receive in order to process the remarks to improve the text. How can you make sure that your feedback suggestions are understandable? These four steps may help with that:

1. Read the review checklist and the text thoroughly. First read the instructions, so that you know what to pay attention to in the text. Then, read the text. First scan it, to get an idea of the structure, then start reading in more detail.
2. When you write a peer review, you are often asked to first give a brief summary of the study at hand. That way, as reviewer you show that you have really read and understood the text. Furthermore, having to summarize forces the reviewer to really look into the cohesion of the text: how do all parts relate to one another? Do the conclusions follow logically from all the previous materials?
3. Identify reading problems. Notice your eye movements while reading. For example: your eyes may get stuck on a sentence. Is this because you're tired? Then take a break. But it may also be that the sentence is unclear. In that case, make a note that you had a reading problem there and continue reading. Get back to that passage later to find out what exactly you did not understand about the text.
4. Explain what the problem is. Why is the text not clear to you? Is it because you miss background information? Is it because the reasoning is unclear? Is the structure illogical? Or are there local problems, such as a grammatical error or a spelling mistake. Global problems with the content and the structure are more worthwhile to pay attention to than grammatical errors or spelling mistakes: they lead to more important revisions. Of course it is important to point out local problems as well, but do not pay too much detailed attention to these. Give a few examples of the spelling errors you encountered, but leave it at that.
5. Finally, make concrete suggestions for improvement. the more concrete your suggestion is, the clearer it is. Try to explain what your solution to the writing problem would be, if you were the author.

1. Checklist

What needs attention	What is expected from you?	What goes well
	Elements	
Since there is no abstract yet, the RQ only appears in two places (page 2 and the introduction). In the final version it should appear consistently in the abstract, introduction, and conclusion. Right now the conclusion section does not restate the research question before jumping into the findings. It would help the reader if you opened the conclusion by referring back to what you set out to investigate.	Objective/Research question/Aim • is presented consistently in Abstract, Introduction and Conclusion • is specific enough • can be answered with a literature study	The research question is clear and works well for a literature review: "How do different reconfigurable intelligent surface configurations affect communication link strength?" It is specific enough to guide the paper without being too narrow. The sub-questions on page 2 break the main question down into logical parts, which is helpful. The RQ also comes back in the last paragraph of the introduction, so the reader knows what to expect.
The introduction tries to cover too much ground. It already discusses passive vs. active trade-offs, hybrid configurations, RF energy harvesting, and beamforming in quite some detail. This makes it read more like a summary of findings than an introduction. I would suggest moving the detailed passive/active comparison into the body sections and keeping the introduction focused on the problem and the RQ. There is also no sentence that tells the reader how the rest of the paper is organized. Something like "This review first compares the simulation approaches used across studies, then presents their results, and concludes with a discussion of the implications" would help a lot. The "Structure" section on page 2 looks like it was carried over from Assignment 1. This should not be a separate section. Instead, work a brief structure overview into the end of the introduction.	Introduction • contains background information on the problem • indicates a motivation for the research / points out the problem in the present situation or previous research and argues why this issue should be resolved • introduces all relevant concepts, such that a reader can understand all components in the research question sufficiently • contains an objective (in the form of a(n) aim/purpose statement/hypothesis/...) • indicates the approach/method (criteria used for comparison, search engines used, how the sources were filtered, etc.) • indicates the structure of the article / description of how this paper answers the research question (unless AIMRaD has been used).	The introduction gives a clear picture of why RIS technology matters: the move toward 6G, the problems with mmWave frequencies, and the limitations of treating the wireless channel as uncontrollable. The explanation of what an RIS actually is and how passive and active versions differ is written in a way that someone outside the telecom track can follow. The motivation for the review (that there are few comparative analyses of how RIS architectures affect link strength) is convincing.
The main issue here is that Section 2 compares the methods used in the reviewed papers, but the paper is missing a methodology section for the literature review itself. The assignment requires you to explain how you searched for sources: which databases you used, what search terms you tried, what time range you set, how you decided which papers to include or exclude, and how many you found vs. how many you kept. Without this, a reader cannot tell whether the 5 sources represent a thorough search or just the first few papers that came up.	Method (if not mentioned in Introduction) • explains which method was used to find suitable sources for the literature • explains how the results were filtered • contains the criteria used for comparison • is formulated in such a way that it is replicable	Section 2 does a nice job of comparing the simulation and experimental setups used in the different papers. The comparison between Radpour et al. and Amri et al. (different element counts, element sizes, and frequencies) is useful, and pointing out the limitations of their setups (indoor only, blocked direct link) shows that you are reading the papers critically, not just summarizing them.

I would suggest adding a short methodology section (around 100-200 words) between the introduction and the current Section 2. You could also rename Section 2 to something like "Comparison of Simulation Setups" to avoid confusion with the methodology.		
The biggest issue is that the body only uses 5 sources, while the assignment requires at least 9. This is something that really needs to be fixed before the final submission. You could look into papers on RIS hardware prototypes, channel estimation, deployment strategies, or comparisons with relay-based systems. There is a typo in the Section 3.2 heading: "Benifits" should be "Benefits." The paragraph at the bottom of page 5 starting with "In conclusion, the evidence from these studies..." reads like a conclusion, but it appears before Section 4 (Conclusion and Discussion). This is a bit confusing because it feels like you are concluding twice. I would move that paragraph into Section 4 and remove it from the results section.	Body text - contains a full account of all results of the literature review. - has clear headings that show the storyline.	The body has a clear structure with useful headings: Active RIS results, Adaptive Beamforming, mmWave operation, and a summary table. Table 3 is especially effective because it lets the reader quickly compare the key metrics across all four main sources. The subsections follow a logical order and each one builds on the previous one.
Reference use		
As mentioned above, only 5 sources are used and the minimum is 9. This needs to be addressed. There are also a few claims in the introduction that do not have any citations. For example, the sentences about hybrid RIS frameworks, RF energy harvesting, and IoT applications are not backed up by references. Either add citations for these or consider removing the sentences if they are not central to your argument.	Use of sources - sources are relevant and trustworthy - when ideas, fragments, figures and data have been quoted from other work, they are accompanied by correct in-text citations - literal quotations have quotation marks and page numbers - paraphrases are not too close to the original	The sources are used well throughout the text. Claims are backed up with specific numbers (SNR values, BER reductions, sum-rate gains), and you compare findings across sources rather than just describing each one separately. That is good synthesis. Three of the five sources are from 2024 or later, which meets that part of the requirement.
The reference format does not follow IEEE style. For example, IEEE would format reference [2] as: M. M. Amri, N. M. Tran, and K. W. Choi, "Reconfigurable intelligent surface-aided ..., " IEEE Access, vol. 9, pp. 147442-147457, 2021. The current format uses "In:" before journal and conference names, which is not how IEEE does it. I would recommend using the IEEEtran bibliography style in LaTeX, which handles the formatting automatically. Also check that all entries are consistent (some list all authors, others use "et al.").	Bibliography - the references and bibliography meet the requirements (IEEE-style) - every in-text citation in the text corresponds to a reference in the bibliography and vice versa	The references are numbered and each in-text citation matches an entry in the bibliography. The entries contain the essential information (authors, title, publication venue, year, and pages).
Argumentation & Structure		
Some paragraphs jump straight into describing a specific study without first telling the reader what the paragraph is about. For instance, the	Paragraphs are well-constructed	The paragraphs in the results section are generally well structured. Each one focuses on a single study

paragraph in Section 2 that starts with "Radpour et al. simulated an RIS-panel..." could use a topic sentence like "Several studies have used simulation to evaluate RIS performance in indoor scenarios, though their setups differ in important ways." The transitions between sections could also be smoother. The jump from Section 2 to Section 3 feels a bit abrupt. A sentence connecting the two (e.g., explaining that now you will look at what these different setups actually found) would help the reader follow the storyline.	- are not too long: each paragraph discusses one topic - contain a clear topic sentence showing the topic of that paragraph and a link to the previous paragraph - are divided into a clear paragraph structure	or comparison point, which keeps things organized. The paragraph in Section 3.3 that compares Radpour et al. and Enahoro et al. at mmWave frequencies is a good example of how to bring two sources together to make a point.
The "Research question," "Structure," and "Statement on the use of generative AI" sections on page 2 interrupt the flow of the paper. They look like they belong to Assignment 1 rather than the actual review. The AI statement should go in an Acknowledgement section at the end of the paper. The paragraph at the bottom of page 5 that starts with "In conclusion..." appears before the actual conclusion section. This weakens the impact of Section 4 because the reader has already read something that sounds like a conclusion. Consider moving it into Section 4. Also, the structure on page 2 lists "Conclusion" before "Discussion," but usually discussion comes first (you reflect on findings, then conclude). You might want to rename Section 4 to "Discussion and Conclusion."	Storyline - there is a clear storyline/main theme throughout the paper - the conclusion matches the Introduction and the intermediate sections lead the reader from Introduction to Conclusion.	There is a clear overall structure: the paper explains why RIS matters, then compares how different studies approach the problem, then looks at what they found, and wraps up with a conclusion. Table 3 does a good job of tying the results together before moving to the conclusion.
The critical analysis could go a bit deeper in places. For example, Zhang et al. [4] report a 10,000x SNR advantage for active over passive RIS, but this is based on a simulation with specific parameters (Table 2). It would be worth asking how realistic those parameters are and whether the gains would hold under different conditions. It would also strengthen the paper to discuss limitations of the review itself. Since you only use 5 sources and all the results come from simulations, the reader should know that real-world measurement data is still scarce, which limits how strong the conclusions can be. Not all sub-questions from page 2 are clearly answered in the body. For example, "How do the physics behind different RIS configurations work?" is only partially addressed.	Argumentation - the argumentation is clearly recognizable - is clear and well structured - reasoning is built upon sources - contains a critical comparison of multiple sources	The paper makes a clear case that active RIS outperforms passive RIS in terms of SNR and throughput, but comes with higher power consumption and complexity. This argument is supported with concrete numbers from multiple sources (the tables are very helpful here). The observation that beamforming algorithms are just as important as the hardware for passive RIS performance adds useful nuance.
Style		
There are a few errors that should be fixed: - Section 3.2 heading: "Benifits" should be "Benefits" - Page 4: "proving control" should be "providing control"	- correct use of English: understandable, no spelling errors, acceptable grammar, no lexical mistakes - the style used is appropriate to an academic register:	The writing is generally clear and easy to follow. Technical terms like double-fading effect, beamforming, SNR, and BER are explained well

- Page 5: "simulations shows" should be "simulations show" - Page 5: "evidence from these studies suggest" should be "suggests" - Page 5: "that that beamforming" has a duplicate "that" Some sentences in the introduction try to pack too many ideas into one sentence. Splitting them up would make the text easier to read.	- objective, relatively formal and concise - the writing style is clear and reader friendly: sentences are not too long or complicated and there is no ambiguity.	enough that someone from a different EE track can understand them. The tone is appropriate for an academic paper.
Illustrations and layout		
There are no figures in the paper. A diagram showing how passive and active RIS architectures differ, or a visual of the simulation setups, would make it easier for readers who are not familiar with the topic to understand the concepts. The caption for Table 2 ("Parameters used in [4] to asymptotically evaluate SNR performance") could be more descriptive. Something like "Simulation parameters for comparing passive and active RIS SNR performance" would give the reader a better idea of what they are looking at.	Tables and figures - all have a number and a caption - all are referred to at least once in the text - if copied, they contain a reference - can be interpreted on their own (e.g. by means of a legend)	The paper includes three tables, all with numbers and captions. Table 3 in particular is very useful as a summary comparison. The tables are all referenced in the text.
The paper is not formatted according to IEEE author guidelines. It uses a single-column layout instead of the required two-column IEEE conference format. This is something that needs to be fixed. There are also several required elements that are missing: - Student numbers next to the author names - Footnote on the first page with the course code (EE4C01) and word count - An abstract (150-250 words) - The title and authors should not be on a separate page. In IEEE format, everything starts on the same page - The AI use statement should be moved to an Acknowledgement section at the end	Format: - functional font (type and size) - headings are easily recognizable - professional page lay-out	The headings and subheadings are easy to spot and the text is well spaced.
The paper covers an interesting topic and the research question is well chosen. The strongest parts are the quantitative comparisons between passive and active RIS (especially Table 3) and the critical discussion of simulation setup limitations in Section 2. The writing is generally clear and the structure makes sense. There are a few things that need to be fixed before the final submission: 1. The paper needs to be reformatted to IEEE two-column style, with all the required elements (footnote, student numbers, abstract, acknowledgement for AI use). 2. The number of sources needs to go from 5 to at least 9. 3. An abstract needs to be added (150-250 words). 4. A proper methodology section for the literature review is needed (which databases you searched, what keywords, how you filtered). 5. The sections on page 2 (Research question, Structure, AI statement) should be removed or integrated properly into the paper.		

The foundation is solid. Fixing these issues will make a big difference for the final version.